

TARANDEEP SINGH JUNEJA

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Summary

Final-year B.Tech Computer Science student at VIT with strong foundations in **Machine Learning**, **Computer Vision**, and **Python-based systems**. Experienced in building ML-driven applications, training and deploying models, and developing scalable inference APIs through projects and a **Python Developer Internship at IIT Bombay (FOSSEE)**, with **Global Rank 186 in Meta Hacker Cup**.

Experience

FOSSEE Summer Intern – Python Developer

Feb 2025 – Jun 2025

IIT Bombay

Mumbai Internship Report

- Developed Python-based computation and validation modules for **Osdag**, gaining experience in production-grade code, testing, and large-scale system reliability.
- Worked with structured logic, debugging, and SDLC practices to ensure correctness and maintainability of engineering software.

Software Taxonomy Analyst (Contractual)

Sep 2024 – Jan 2025

Outlier AI

Remote

- Annotated and analyzed **100+ multi-language code snippets (Python, C++)** to support ML-based code understanding systems.
- Applied structured reasoning and edge-case analysis to improve dataset quality and label consistency.

Projects

DherScaler AI — Distributed RAG System [\[GitHub\]](#)

2025

- Developed a **Retrieval-Augmented Generation (RAG)** system by extending the Haystack framework with **Ray** for distributed processing, reducing inference latency by approximately **50%** during large-dataset evaluations.
- Deployed the RAG pipeline as a **REST API using Ray Serve**, enabling scalable multi-node query handling with monitoring and error-handling for robust inference workflows.
- Customized retrieval components with **Chroma vector databases** and text-to-SQL style querying, ensuring reproducibility using **Docker** and **Git**.

DocuGenAI — AI-Driven Document Generation Platform [\[GitHub\]](#) [\[Live\]](#)

Sep 2025

- Built an AI-powered document generation system using **FastAPI**, integrating generative pipelines for dynamic content creation.
- Implemented asynchronous processing and fallback orchestration to ensure high availability and low-latency responses.

Shodh Saathi — AI-Powered Research Automation Platform

2025

- Developed an **AI-powered research automation platform** using **Python**, **LangChain**, and **Google Gemini API** to aggregate information from multiple sources including Wikipedia and DuckDuckGo with **type-safe Pydantic models** for structured data processing and JSON/Markdown exports.
- Built a responsive **full-stack web application** using **Streamlit**, featuring real-time analytics dashboards, custom CSS styling, auto-save functionality, and local storage with downloadable research outputs.

CNN Audio Classifier — PyTorch, FastAPI [\[GitHub\]](#)

Jul 2024

- Trained a **CNN-based audio classification model** on the ESC-50 dataset using Mel-spectrogram features, achieving **83.5% accuracy**.
- Built a FastAPI-based inference service to expose predictions via RESTful endpoints.

Research & Exploratory Work

- Conducted comparative experiments on object detection models by tuning augmentation strategies, confidence thresholds, and IoU values to study their impact on detection performance.
- Performed error analysis on misclassified samples to identify dataset imbalance and annotation issues, improving evaluation reliability.
- Documented experimental findings and trade-offs between accuracy, latency, and deployment feasibility for applied ML use cases.

Education

Vellore Institute of Technology

2022 – 2026

B.Tech Computer Science and Engineering

CGPA: 8.54 / 10

Technical Skills

Programming & Core CS: Python, C++, SQL, Data Structures and Algorithms, Object-Oriented Programming, Operating Systems

Machine Learning & Computer Vision: PyTorch, Scikit-learn, Convolutional Neural Networks, YOLOv8, Model Training, Evaluation, Deployment

Tools & Platforms: FastAPI, Docker, Git, GitHub, Postman, Roboflow, CI/CD Pipelines

Achievements & Certifications

- Meta Hacker Cup 2025 — Global Rank **186** (Top **0.5%**)
- Oracle OCI Generative AI Professional — 2025
- Postman API Student Expert — 2024
- Omdena Contributor — Machine Learning & Computer Vision (Global Collaboration)

Coding Profiles

700+ problems solved across [Codolio](#) — [LeetCode](#) — [GeeksforGeeks](#) — [CodeForces](#)